SOFTWARE ENGINEERING 2: PROBLEM PROJECT PROPOSAL

Research Group Dissertation & Presentation

Client: Amenda Raganya

Group Allocation: Group 3 – Cape Town Campus

Group Members: Willem Basson, Dylan Shepard, Moegammad Green, Sihle Kayise

Submission Data: 2 October 2026

Selected Group Lead

- **Group Lead:** Willem Basson
- **Coordination Role:** Responsible for client communication via Microsoft Teams, milestone tracking, repository management, and consolidating dissertation deliverables across the four members.

Problem Statement

- **Domain:** Higher Education Administration & Workplace Compliance
- **Background & Context:** Occupational qualifications require workplace hours, workshops, and portfolios before the final external assessment. Tracking still uses PDF forms, emailed spreadsheets, and physical sign-offs.
- **Core Problem:** Entries are lost or unverified, at-risk students are noticed late, coordinators have no live view, and audits are slow. An LMS grades classwork and does not verify host employers. Dedicated placement systems (Chapter 2) assume a full university rollout.
- **Proposed Software Engineering Solution:** TrackWIL, a role-based web app. Core product: logbook entry, secure token verification, and a coordinator dashboard. Analytics, bulk exports, and audit logs are stretch goals, and only if the core is finished before 29 October 2026.
- **Research & Data Feasibility:** Questionnaires, time-on-task comparisons, and usability tests with staff and students. No personal information is collected until written campus permission, and it is protected under POPIA.

Stakeholders, Core Product, And Stretch Goals

- **Students:** Log tasks, upload evidence, and see hours completed.
- **Host Employers:** Sign off from a secure link, with no account.
- **Coordinators:** See on-track, awaiting verification, and overdue logbooks.
- **Core:**
    - Logbook entry with evidence and hour progress.
    - A Time-limited token so the employer verifies without an account
    - The coordinator dashboard.
- **Stretch, only after the core works and only before 29 Oktober:** Analytics beyond the dashboard, bulk compliance exports, and an audit log.

Chapter 2: Existing WIL and Placement Tools

Record hours, have the host verify them, and show the coordinator who is behind. Existing tools already do this. TrackWIL keeps those three jobs and removes the account and license burden on the host employer.

Tool	Logbook	Employer check	Coordinator view	Limit for this Project
PDF, email, paper	Form or sheet	Physical or Email sign-off	Manual collation	Current process; Entries get lost
Module (LMS)	Assignment upload	Needs an enrolled account	Gradebook	Built for classroom grading
InPlace	Student Log Book	Supervisor portal account	Portal and analytics	Licensed university WIL suite
SONIA	Timesheets	Mentor account	Placement console	Built for a placement office
PebblePad	Workbook activity log	External assessor access	Workbook reports	Portfolio, not hour compliance
TrackWIL	Entry and evidence	Token link, no account	Status and overdue flags	Core product for this window

InPlace, SONIA, and PebblePad show that digital sign-off is established practice, and that it comes with licences, portals, and a mentor account. This campus is still manual, so TrackWIL delivers the three core jobs by 29 October 2026. Analytics, bulk export, and audit history stay on the stretch list.

Methodology

Hybrid Agile-Scrum and a 4+1 view. The week 1–7 plan ran past the window and is replaced below. Build is 3-tier: relational database, REST API with role-based access, responsive web frontend. GitHub holds the code and board. Nothing is scheduled after 29 October 2026.

- **2-8 October – Requirements.** Confirm core and stretch. UML use cases and activity diagram.
- **9-15 Oktober – Sprint 1.** Logbook entry: create, view, evidence, hours completed.
- **16-22 Oktober – Sprint 2.** Expiring token, employer sign-off, coordinator dashboard. Stretch work only if the core is already demonstrable.
- **23 – 29 Oktober – Test and Submit.** Tests of the three core features. SUS and time-on-task only with consent after permission; otherwise, synthetic data, stated in the report. Dissertation, including this chapter, and presentation.

Personal Information (POPIA)

Written campus permission comes before any staff or student data is collected. Topic approval is not that permission. Participation is voluntary; people are told the purpose and that only the group and the supervising lecturer see the data. Only what the core evaluation needs is kept, in access-controlled storage. Demonstrations use synthetic data and tokens expire. Personal information is deleted after the result is finalized, unless the campus sets another period. If permission is late or refused, the build uses synthetic data and the dissertation says so.